Assignment 4 — Production Plan & MVP 2.0 (Alpha Build)

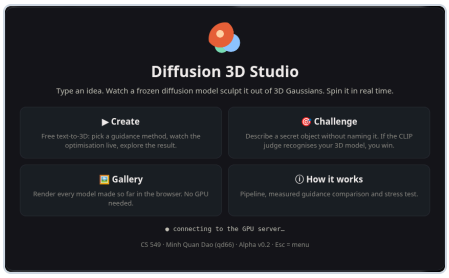
Minh Quan Dao (qd66) · Interactive Diffusion-Based 3D Generation Studio · Pillars: **Procedural Generation + Rendering** · Alpha **v0.2.0**
Playable build: live studio (lab A100 server) · static web build (gallery + viewer, no GPU) · **Build zip:** v0.2.0-alpha.zip · **Source:** github.com/quandao10/diffusion-3d-studio

1. MVP 2.0: from tech demo to a complete loop

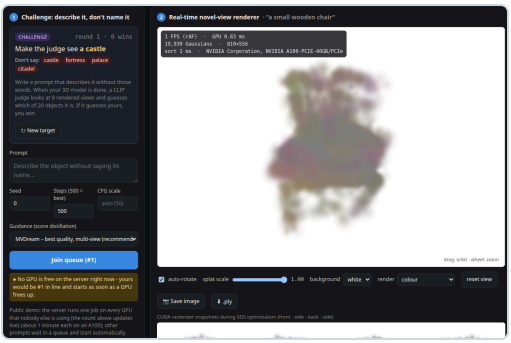
Assignment 3 proved the core mechanic: a text prompt optimises 3D Gaussians with gradients from a frozen diffusion model, and the result is rendered in real time. The Alpha wraps that mechanic in a game loop with a goal, a judge and a win/loss state:

Start: main menu. Create, Challenge, Gallery and How-it-works, plus a live line showing free GPUs, the queue and the time per model.	**Active: the studio.** *Challenge* gives you one of 20 secret objects; you must describe it **without its taboo words** (checked in the browser and on the server). The model is optimised live with ETA and Stop. *Create* is free generation.	**End: results.** A CLIP judge looks at 8 rendered views and ranks the 20 objects: **win** if it picks your target, **loss** otherwise. Then *Play again / Retry / Explore in 3D / Download .ply / Menu*; session score and streak in the HUD.

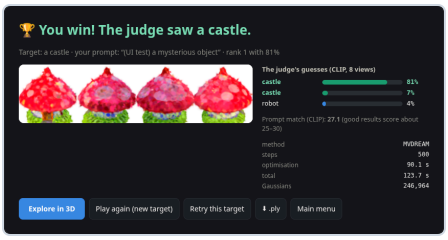
Both pillars are integrated in the loop: procedural generation *is* the gameplay (your words → diffusion-guided Gaussians), and rendering is how you play and are judged (real-time splatting viewer with colour/depth/centre modes; the judge scores rendered views). The judge picks the right object for **49/50** existing results (`evidence/judge_eval.csv`); its only miss is a genuinely broken model.



Start state: main menu.



Active state: Challenge target, taboo words, queue/ETA (all GPUs busy at capture).



End state: result screen (layout shown with UI-test judge values).

2. Release notes (changes since the Assignment 3 midterm)

New features

- **Game loop:** main menu (start), Create and Challenge modes (active), result screen with win / loss / failed / stopped states (end); Esc returns to the menu.
- **Challenge mode:** 20 secret objects with taboo lists (`configs/challenge.json`), live taboo check, *New target*, session rounds / wins / streak.
- **CLIP judge in the pipeline:** after every job, ViT-L/14 scores 8 views: prompt match plus a 20-way zero-shot guess, shown as bars on the result screen.
- **Rendering:** viewer modes colour / depth / Gaussian centres, PNG screenshot, `.ply` download; the app opens on a curated MVDream model.
- **Server:** model-availability check (methods whose weights are missing are disabled with a clear message); the ETA includes the judge.

Bug fixes

- **Jobs could hang for minutes or forever:** every job checked the Hugging Face Hub online at load, and a cache clean-up had deleted MVDream's 5.2 GB checkpoint, so jobs silently re-downloaded it. Jobs now run offline (`HF_HUB_OFFLINE`) from local weights in a project `weights/` folder; SD now loads in 4.8 s.
- **Watchdog:** a job whose log is silent for 5 min is stopped, its GPU released, and the reason shown to the player.
- After a server restart, the page now reports a lost job instead of polling forever; a removed or broken model no longer leaves "loading…" on screen.
- The job panel showed the previous job's numbers when a new job started; it now resets. The viewer recovers from WebGL context loss instead of going blank.

Known issues

- Jobs only use GPUs with *no* other process. At release time all 16 lab A100s (homer and hermes) were busy with other training, so the live build queues jobs until a GPU frees. A real GPU Challenge round is therefore still to be

recorded; the loop is covered by 11 server tests and 14 browser UI tests.
- The CLIP judge can be fooled: it rewards colour and texture, knows only 20 objects, and over-saturated models can be misread.
- MVDream results are still over-saturated for some prompts; the SDS method still makes two-faced objects.
- The quick-tunnel URL changes when the tunnel restarts; the static build cannot generate. "One job per session" relies on `sessionStorage`, so a new tab counts as a new session.

3. Weekly production schedule (Fall 2026, weeks counted from Sep 1; final deadline assumed in week 15)

Week	Engineering	Assets / content	Integration / milestone
9 · Oct 26–Nov 1	**Alpha (done):** game loop (menu → Create/Challenge → result), CLIP judge, render modes, offline jobs, watchdog	20 challenge targets + taboo lists; judge calibration on 50 results	Alpha v0.2 live + tagged (this report)
10 · Nov 2–8	Stage-2 mesh refinement (`main2.py`) + mesh/Gaussian toggle in the viewer; reserve one GPU for the studio so jobs never wait behind other users	Reference render per challenge target (MVDream) for the result screen	Mesh download on the result screen
11 · Nov 9–15	Quality: MVDream + ISM-style deterministic targets; CFG/timestep schedule sweep; judge robustness (prompt ensembles, calibrated threshold)	Expand Challenge to 40 targets with easy/medium/hard tiers	Re-run guidance benchmark; difficulty levels live
12 · Nov 16–22	Persistence: SQLite for jobs, scores and a moderated leaderboard; admin hide/delete for the gallery; stable hosting (named tunnel or CS proxy)	UI polish: onboarding tips, icons, empty/error states	**Integration:** leaderboard + moderation + hosting together; classmate play-test
13 · Nov 23–29 (Thanksgiving)	Viewer performance: GPU sort / level of detail, mobile layout; fix play-test issues	Curate final gallery content	Play-test feedback triage
14 · Nov 30–Dec 6	**Feature freeze (beta):** fix the top issues only; load test with parallel players	Record final demo video	Release candidate v1.0-rc; final report draft
15 · Dec 7–13	**BUFFER WEEK:** bug fixing, documentation and polish only; **no new features**	Final screenshots	v1.0 release + final submission

green = done · purple = integration weeks (parts come together in week 12, frozen in week 14) · orange = buffer week (bug fixes, documentation, polish; no new features).

Run it: `bash scripts/setup_env.sh`, then `bash scripts/serve_public.sh` (or `python server.py` for local use). Linux, NVIDIA GPU with ≥ 12 GB; see the README.